

SEPTEMBER 20, 2017

Exploring Problems with Counter-Strike Rating Systems

When you look at the back of a baseball card, you don't see a single rating. You see time-series information about that player.¹ Additionally, they do not use a single number to tell a story about a player. No magic number that will accurately depict a player's performance. However, with younger sports like Counter-Strike: Global Offensive, the single stat most people focus on is a player's rating. This single number is problematic for many reasons. With the trove of data we have access to in demo files that we do not have with any other sport, we need to move to something more nuanced.²

**TOPPS
50**

MICKEY MANTLE

N. Y. Yankees, of

HT: 6'0"

WT: 200

BATS: BOTH

THROWS: RIGHT

BORN: OCTOBER 20, 1931

HOME: DALLAS, TEX.

Mick was hampered by several serious injuries in '63. The sensational switch-hitter won the A.L. batting crown in '56. He has hit 15 World Series homers.

COMPLETE MAJOR AND MINOR LEAGUE BATTING RECORD

YEAR	TEAM	LEA.	G	AB	R	H	2B	3B	HR	RBI	AVG.
1949	Indep.	K. O. M.	80	323	54	101	15	7	7	63	.313
1950	Joplin	W. A.	137	519	141	199	30	12	26	136	.383
1951	New York	A. L.	96	341	61	91	11	5	13	65	.267
1951	Kansas City	A. A.	40	166	32	60	9	3	11	50	.361
1952	New York	A. L.	142	549	94	171	37	7	23	87	.311
1953	New York	A. L.	127	461	105	136	24	3	21	92	.295
1954	New York	A. L.	146	543	129	163	17	12	27	102	.300
1955	New York	A. L.	147	517	121	158	25	11	37	99	.306
1956	New York	A. L.	150	533	132	188	22	5	52	130	.353
1957	New York	A. L.	144	474	121	173	28	6	34	94	.365
1958	New York	A. L.	150	519	127	158	21	1	42	97	.304
1959	New York	A. L.	144	541	104	154	23	4	31	75	.285
1960	New York	A. L.	153	527	119	145	17	6	40	94	.275
1961	New York	A. L.	153	514	132	163	16	6	54	128	.317
1962	New York	A. L.	123	377	96	121	15	1	30	89	.321
1963	New York	A. L.	65	172	40	54	8	0	15	35	.314
Major League Totals 13 Yrs.			1740	6068	1381	1875	264	67	419	1187	.309

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Recently, HLTV [announced](#) a new rating system for professional CS:GO players. This new system was designed to weight more impactful kills more than less

impactful ones. They describe the system as one that “better encapsulates the different ways players contribute in the game.” Statistically speaking this is untrue. HLTV 2.0 provides no statistically discernible difference that improves it in comparison with its predecessor.

HLTV 1.0 Construction

According to HLTV, the [rating 1.0](#) system uses three distinct elements:

- Kills Per Round
- Survived Rounds Per Round
- A value based on number of rounds with multiple kills

Formula

The formula HLTV uses to calculate the rating can be expressed as follows, where Cs are the respective constants and the summation represents the multi-kill rating:

$$\left(\frac{\frac{K}{R}}{C_0}\right) + 0.7\left(\frac{\frac{R-D}{R}}{C_1}\right) + \left(\frac{\frac{\sum_{i=1}^5 n^2 * k_n}{R}}{C_2}\right)$$

In this context, the Σ calculates an average value for all of the multi-kill events (k_1, k_2 , etc.) weighted by the square of the number of kills. The constants C_n are overall longitudinal averages:

Constant (Average)	Symbol	Value
Kills per Round	C_0	0.679
Survived Rounds	C_1	0.317
Multikill Rating	C_2	1.277

Calculation

This rating system is trivial to calculate in Excel. Since we are given the several constants, leveraging a lookup table allows us to adjust these to determine their

respective effects. These Excel formulas calculate the three variables described above.

Kills Rating

Since this is simple division, we do not need to use anything complicated to calculate a kill rating:

```
=(Kills/Rounds)/0.679
```

Survival Rating

This is similarly simple.

```
=((Rounds-Deaths)/Rounds)/0.317
```

Multikill Rating

This is peskier to calculate, but overall contains only simple addition and division.

```
=((1K+(4*2K)+(9*3K)+(16*4K)+(25*5K))/Rounds)/1.277
```

Combining the Variables

To calculate these all together, simply reference the respective cells and use the Rating 1.0 formula:

```
=(KillRating+(0.7*SurvivalRating) +MultikillRating)/2.7
```

These can also all be combined to a single cell:

```
=((((Kills/Deaths)/0.679)+ (0.7*(((Rounds-Deaths)/Deaths)/0.317)))+  
(((1K+(4*2K)+(9*3K)+(16*4K)+(25*5K))/Rounds)/1.277))/2.7
```

Rating 1.0 Problems

Because rating 1.0 relies on simple mathematics and a small cluster of data, it tends to be biased towards rewarding players with higher kill death ratios. This is problematic because kill death ratio is already an entirely useful metric on its own; having two metrics to judge players that tell you the same thing provides no additional insight.

Reliance on Averages

HLTV 1 relies on three constants. This is problematic for several reasons, not the least of which is that they are not the averages anymore.

Incorrect Averages

Primarily, the averages of 0.679, 0.317, and 1.277 are simply wrong. Looking at the averages we see in the actual statistics; these numbers are closer to the following:

Variable	HLTV Average	Actual Average	Diff
Kills per Round	0.679	0.658	0.021
Survived Rounds	0.317	0.318	0.001
Multikill Rating	1.277	1.193	0.084

These averages shift the actual values of rating 1.0. The HLTV 1.0 column uses the original averages, and the Adjusted column uses the actual averages.

KPR	SPR	MKR	HLTV1	Adj	Diff
0.96	0.37	2.01	1.42	1.47	0.06
0.63	0.38	1.10	0.98	1.01	0.03
0.51	0.36	0.66	0.77	0.79	0.02

Averages are Constants

Additionally, static averages do not provide for comparison of individuals. While using a population mean allows you to compare a player against the overall average, it does not enable one to demonstrate how a player performs relative to their mean. This means the average does not account for map differences, team differences, or other variables that account for variances in kills and deaths.

Reliance on K/D

To demonstrate this problem with rating 1.0, we need only run a simple linear regression to determine the effect the kill death ratio has on individual players. Using the Ratio 1.0 table from my [database](#) of player statistics, we can confidently say that the kill death ratio explains 88% of the change in rating 1.0:

```
. reg rating kd if kd < 3
```

Source	SS	df	MS	Number of obs	=	266,860
Model	26600.3388	1	26600.3388	F(1, 266858)	>	99999.00
Residual	3375.74832	266,858	.01264998	Prob > F	=	0.0000
				R-squared	=	0.8874
				Adj R-squared	=	0.8874
Total	29976.0871	266,859	.112329309	Root MSE	=	.11247

rating	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
kd	.6497441	.0004481	1450.10	0.000	.6488659	.6506223
_cons	.2921196	.0005207	561.00	0.000	.291099	.2931401

Using a Regression Equation

This regression demonstrates that we can use the following formula to estimate rating 1.0 for a player:

$$Rating_1 = 0.2921196 + 0.6497441\left(\frac{K}{D}\right)$$

This is a far cry from the official equation, yet it nets us 88% of the change. If we instead split the regression into kills and deaths instead of a simple ratio, we get an ever more accurate picture:

```
. reg rating kills deaths,r
```

Linear regression				Number of obs	=	270,799
				F(2, 270796)	>	99999.00
				Prob > F	=	0.0000
				R-squared	=	0.9324
				Root MSE	=	.09194

rating	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
kills	.0474294	.000042	1129.55	0.000	.0473471	.0475117
deaths	-.0445878	.0000653	-682.49	0.000	-.0447159	-.0444598
_cons	.9431295	.0012908	730.64	0.000	.9405995	.9456595

This regression allows us to predict rating 1 with even more accuracy, this time with a 5% improvement in our correlation coefficient. The formula for this

regression is as follows:

$$Rating_1 = 0.9431295 + 0.0474294(K) - 0.0445878(D)$$

Increasing R²

If we add HLTV's KAST³ ratio and average damage per round into the mix, the correlation coefficient increases to almost 95%:

```
. reg rating kills deaths kast adr,r
```

Linear regression		Number of obs	=	170,339
		F(4, 170334)	>	99999.00
		Prob > F	=	0.0000
		R-squared	=	0.9412
		Root MSE	=	.08539

rating	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
kills	.041817	.0000814	513.96	0.000	.0416576	.0419765
deaths	-.0413426	.000096	-430.78	0.000	-.0415307	-.0411545
kast	-.0009202	.000028	-32.87	0.000	-.000975	-.0008653
adr	.0028195	.0000246	114.74	0.000	.0027713	.0028676
_cons	.8356406	.0029059	287.56	0.000	.829945	.8413362


```
. vif
```

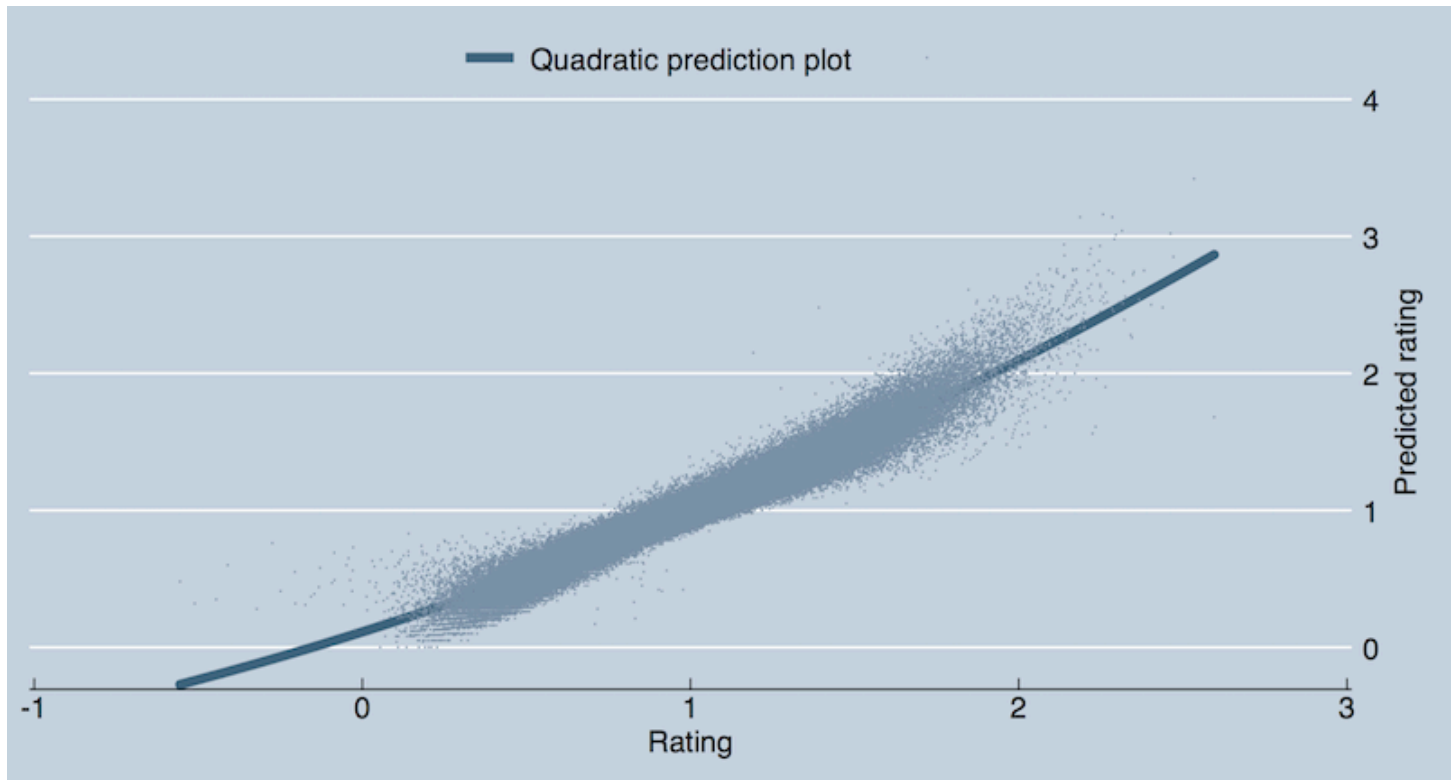
Variable	VIF	1/VIF
kills	3.52	0.283804
adr	3.05	0.327466
kast	2.15	0.465070
deaths	2.03	0.492465
Mean VIF	2.69	

Here, the variance inflation factors demonstrate that there is no multicollinearity problem with the data and that each variable has an effect on the outcome.

Interestingly, here the KAST ratio has a marginally negative effect on a player's rating: the higher a player's KAST is, the more negatively it affects a player's rating.

Predicting Rating 1

Using this to predict HLTV Rating 1 and plotting yields this curve:



Naturally, there is some degree of error, however explaining over 88% of the change in rating 1 with the kill death ratio is problematic because the entire point of rating 1.0 is to *not* be a kill death ratio. Because of these problems, HLTV created their [rating 2.0 system](#) in June of 2017.

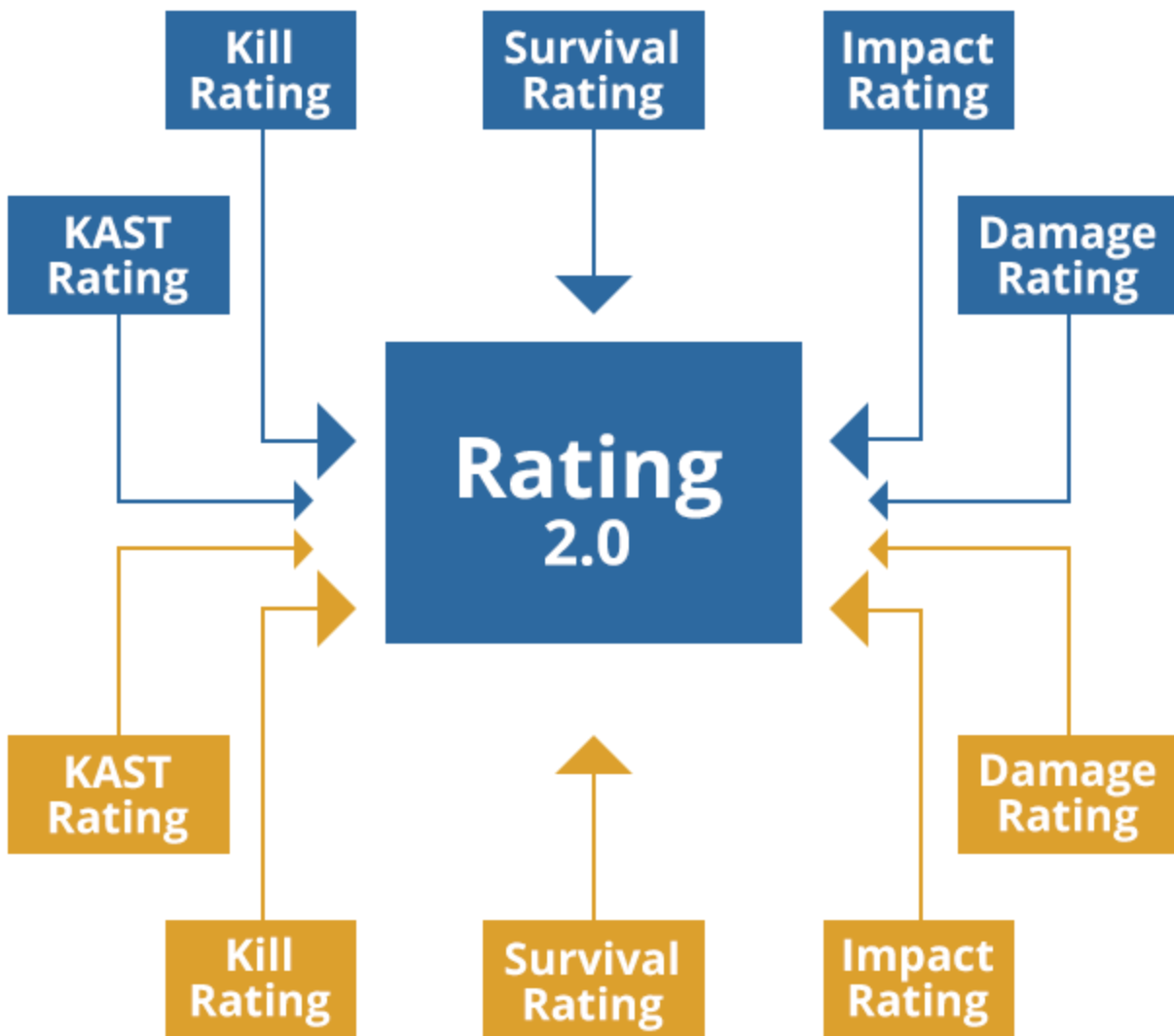
HLTV 2.0 Construction

HLTV describes the rating 2.0 system in their [blog post](#):

[Rating 2.0 is] an updated formula for the Rating which we use to quickly assess player performance. It now incorporates data like damage dealt, opening kills, 1onX wins, traded deaths and more, and thus better encapsulates the different ways players contribute in the game.

Graphically, the inputs are interesting to see. This is a more advanced system that incorporates more variables and attempts to explain more variation in player

performance.



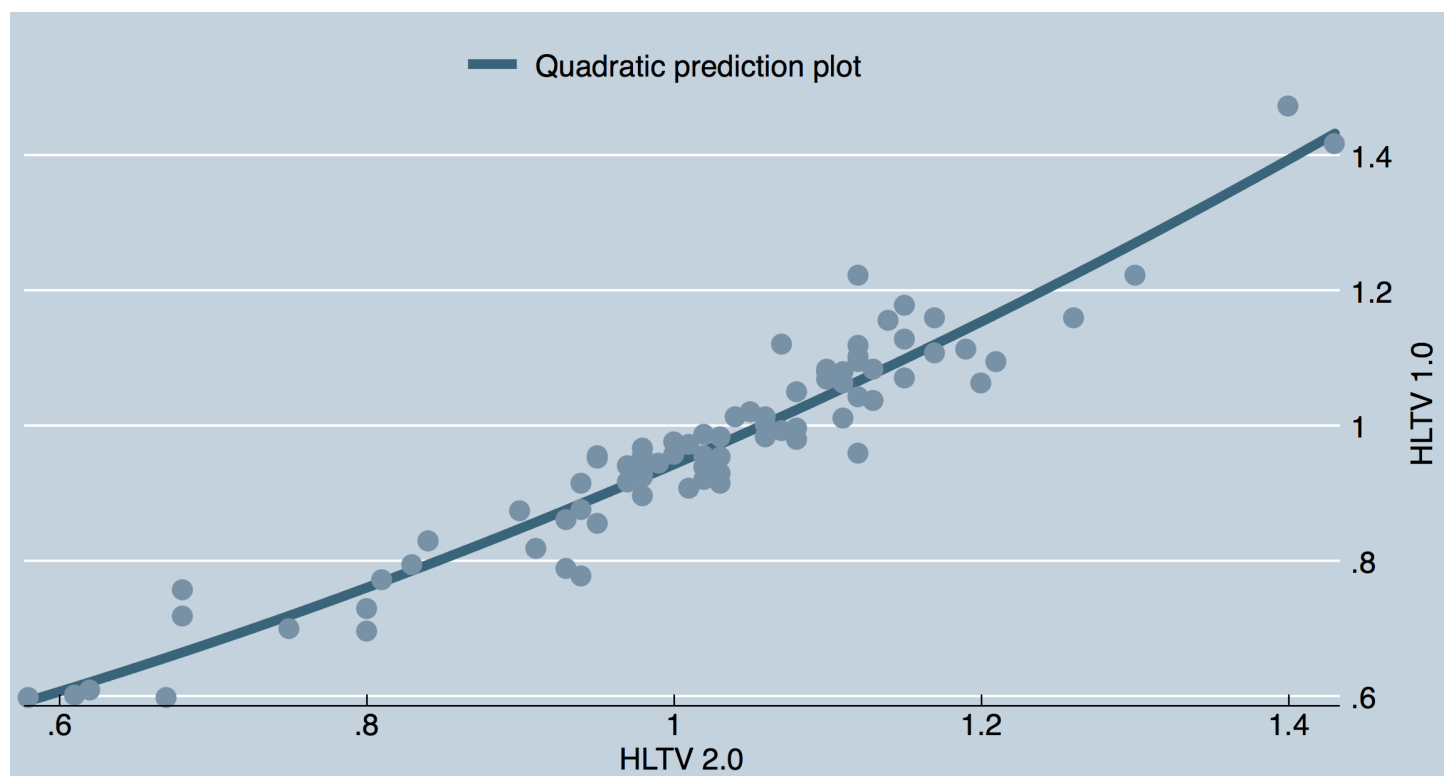
Variables

According to the post, it uses ten inputs instead of the previous three (one set of five for each side played):

- Kill Rating
- Survival Rating
- KAST Rating
- New Impact Rating
- Damage Rating

Comparison

As they explain, this boosts some players and reduces others. For a sample of the 80 players at [PGL Krakow 2017](#), we can see how Rating 1 and Rating 2 compare:



Overall the ratings are quite similar. In fact, their pairwise correlation value is **0.9529**. Unsurprisingly, the two rating systems are highly correlated.

HLTV 2.0 Formula

HLTV decided against releasing the formula for rating 2.0, but this does not mean we cannot use the same type of econometric analysis to predict something very close using the same variables. Using this method we can also compare this new system to the older system by determining how the variables that affected rating 1.0 affect 2.0.

HLTV 2 Problems

As with rating 1.0, there are identical problems with what variables influence the outcome the most.

Testing Kill/Death ratio

To test this variable using the 200,000+ observations in the HLTV dataset, we must first generate the K/D variable.

```
. gen kd = kills/deaths
(8 missing values generated)

. edit

. sum
```

Variable	Obs	Mean	Std. Dev.	Min	Max
map	0				
player	209,324	8135.116	3638.421	2	15300
kills	209,324	17.72975	6.438396	0	91
deaths	209,324	17.76872	4.949734	0	94
adr	209,323	75.62143	19.17653	.6	189.7
kast	209,324	68.97591	12.66756	5.6	100
rating	209,324	1.062465	.352665	.02	3.08
matchid	0				
kd	209,316	1.095448	.6116359	0	26

This also gives us some interesting information about aggregated player performance: the average player has a KAST ratio of 69% while the mean rating is 1.06, slightly larger than the target average rating of 1.0.

Testing K/D

Now that we have the variable we can run the simple regression:

```
. reg rating kd, r
```

Linear regression		Number of obs	=	209,316
		F(1, 209314)	=	4344.44
		Prob > F	=	0.0000
		R-squared	=	0.7391
		Root MSE	=	.18012

rating	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
kd	.4956596	.00752	65.91	0.000	.4809206	.5103985
_cons	.5194738	.0080637	64.42	0.000	.5036692	.5352784

This tells us that KD alone explains 74% of the change in Rating 2.0. While this is down from the 88% of rating 1.0, it does not demonstrate a remarkable shift from the reliance on K/D as a metric for determining player performance.

In fact, when testing separately on Kills and Deaths in lieu of using their ratio, the R^2 increases back to 88%:

```
. reg rating kills deaths, r
```

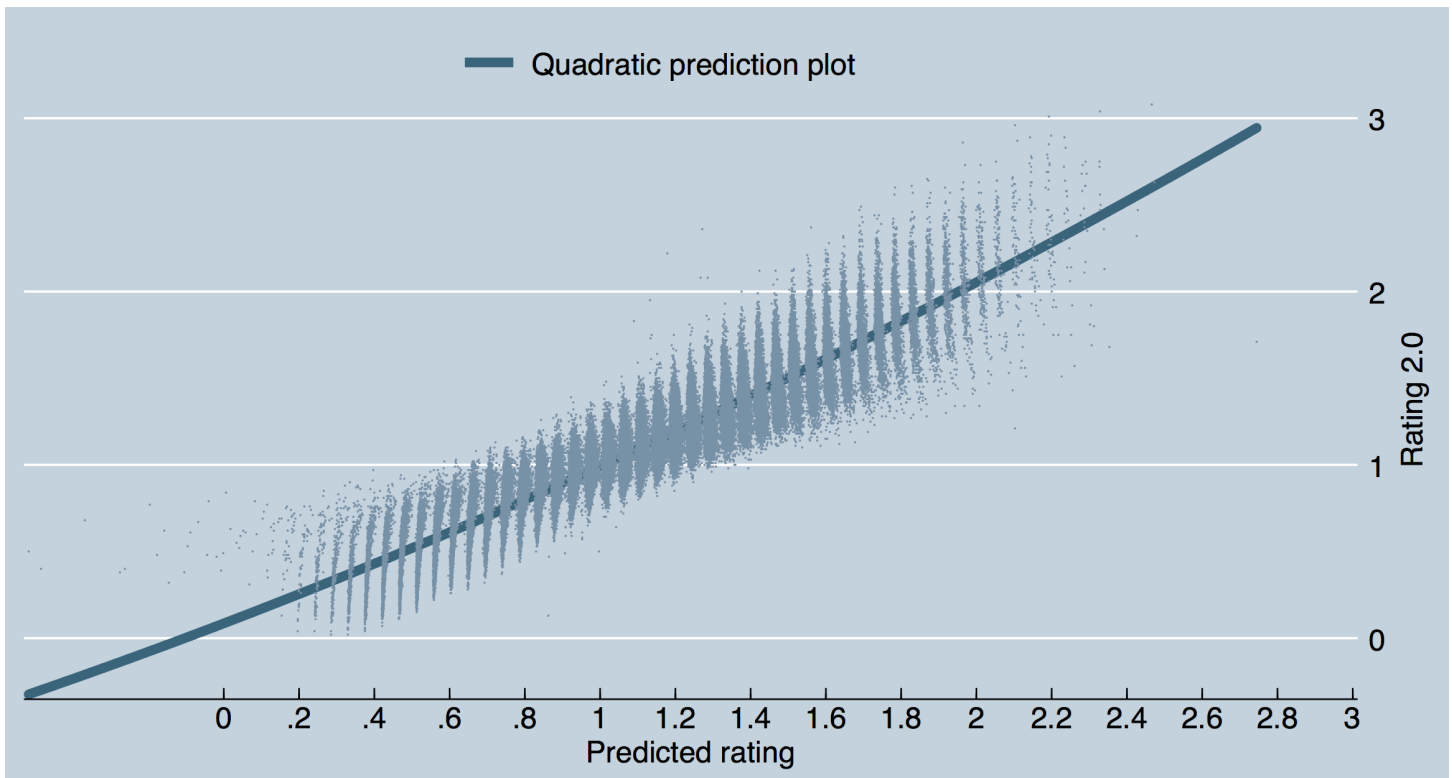
Linear regression		Number of obs	=	209,324
		F(2, 209321)	>	99999.00
		Prob > F	=	0.0000
		R-squared	=	0.8845
		Root MSE	=	.11987

rating	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
kills	.0457674	.0000534	856.48	0.000	.0456627	.0458722
deaths	-.0442967	.0000813	-544.57	0.000	-.0444561	-.0441372
_cons	1.038115	.0016284	637.50	0.000	1.034923	1.041306

This means that HLTV 2.0's reliance does not fall squarely on the K/D ratio per se. However, it is strongly positively correlated with the number of kills a player gets and strongly negatively correlated with the number of deaths a player has.

Prediction Based on K/D

When using this regression equation to predict the value of Rating 2.0 using only Kills and Deaths we see the following trend:



However, with the additional data in the HLTV database, we can increase the accuracy of this prediction.

Adding More Variables

As established with Rating 1.0, ADR and KAST ratio are significant variables that influence rating. Including them in the earlier regression increases the R^2 to 95%:

```
reg rating kills deaths adr kast, r
```

Linear regression

```
Number of obs    =    209,323
F(4, 209318)     >    99999.00
Prob > F         =     0.0000
R-squared        =     0.9565
Root MSE        =     .07358
```

rating	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
kills	.0227886	.0000549	414.84	0.000	.0226809	.0228962
deaths	-.0255902	.0000645	-396.88	0.000	-.0257166	-.0254639
adr	.0065136	.000019	342.15	0.000	.0064763	.0065509
kast	.0063326	.0000202	314.15	0.000	.0062931	.0063721
_cons	.1837702	.0020284	90.60	0.000	.1797946	.1877458

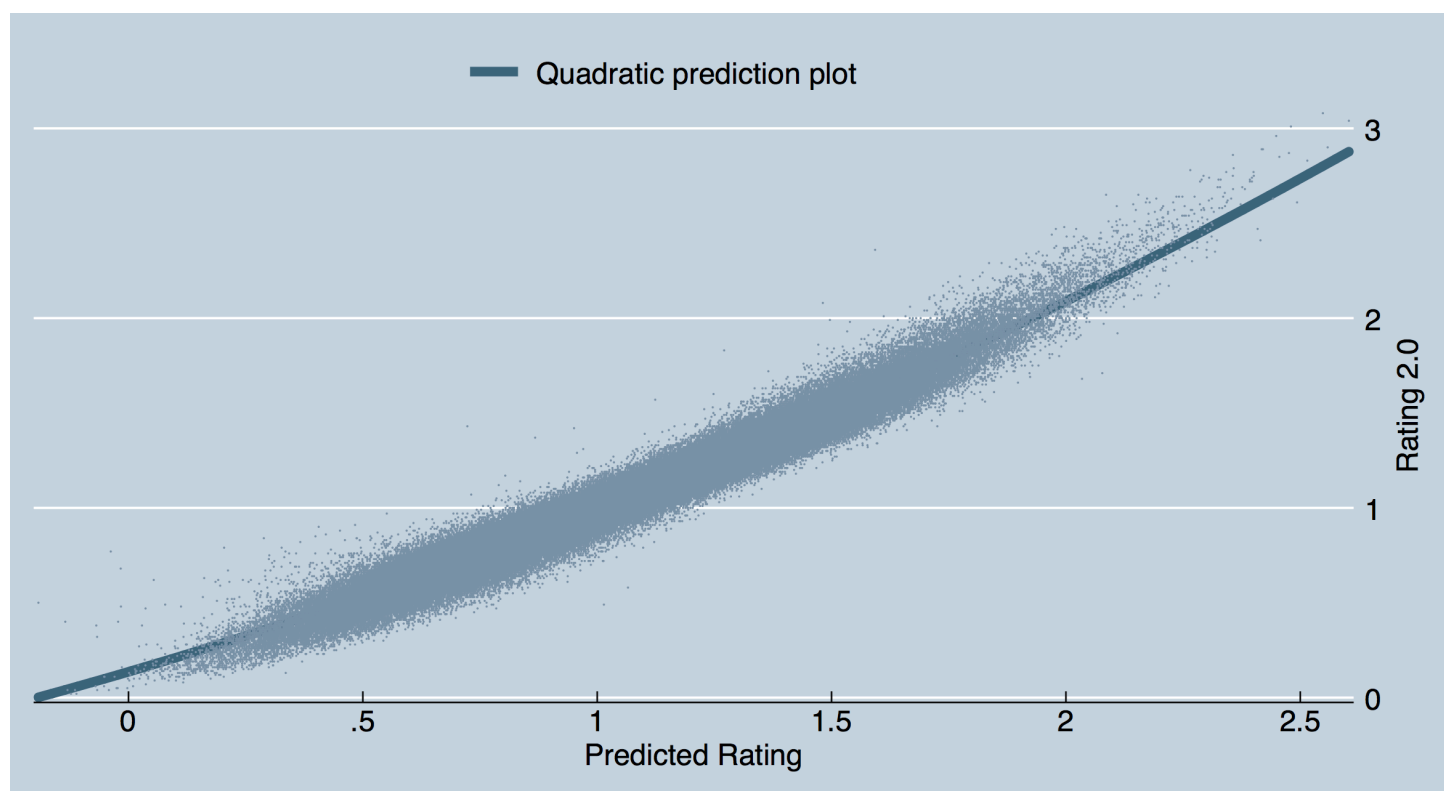
This regression demonstrates that all variables are statistically significant in influencing the value for Rating 2.0. It also evinces that all variables are positively correlated with Rating 2.0 aside from deaths, as we expect. This is different from Rating 1.0 which has a marginal negative correlation to the KAST ratio.

Thus, we can use this formula to predict the value:

$$\text{Rating2} = (0.2778 * \text{Kills}) + (-0.2559 * \text{Deaths}) + (0.00651 * \text{ADR}) + (0.00633 * \text{KAST}) + 0.18377$$

Plotting the New Prediction Formula

Using the above formula we can plot the predicted rating against the actual rating. This correlation is far stronger at 95% R^2 and lacks the binning pattern in the original rating 2.0 prediction:



Conclusions

These regressions demonstrate that rating 2.0 simply plots the K/D ratio with little else thrown in to change the number. This does not mean that HLTVs claims that they use ten different pieces of information to calculate the new rating system. However, it does mean that we can arrive at the same conclusion with fewer, less complicated variables.

Improvements

As a result, we should take rating 1.0 and 2.0 with a grain of salt. The other four statistics that HLTV has on their page are far more telling about different aspects of player performance than these rating systems. Additionally, trade ratios should be accounted for.⁴

Trade ratios are particularly important as they are not only a measure of team play but also a measure of individual skill: should a player have a high ADR or K/D and low trade death ratio, the numbers evince that player's aggression has paid off. Likewise, low traded deaths and low K/D would reveal that the same aggression is not paying off. Additionally, entry kills and holds are important factors for determining whether teams can hold a site or enter a site.

As with baseball, it does not make sense to compare pitchers to batters. Similarly, it is not logical to compare riflers, AWPers, entry fraggers, and IGLs. There are ways to measure each of these things distinctly—and we should do so.

Discussion: [r/GlobalOffensive](#)

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1. [Example here.](#) ↩
 2. Developing new rating systems is something I will address in a future blog post. The purpose of this post is to demonstrate *why* we need new rating systems. ↩
 3. KAST stands for rounds with a Kill, Assist, Survival, or Trade. ↩
 4. As I have in my [prior analyses.](#) ↩

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